

SCHIZOPHRENIA

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INTRODUCTION

Importance

- Schizophrenia is a severe psychotic disorder characterized by disturbances in cognition, emotional responsiveness, and behaviour.

Epidemiology

- It affects 1 in 100 persons ¹. Males and females are affected with equal frequency.
- Common ages of onset:
 - Males: 15-25 years old
 - Females: 25-35 years old
 - Another peak around 40+ years old.
- The risk of suicide is 1 in 10.
- Schizophrenia can result in significant burden to the individual. Patients with schizophrenia experience a higher risk of suicide, substance abuse, and violence compared to the general population.

DIAGNOSTIC CRITERIA ACCORDING TO DSM 5

- Two or more of these symptoms must be present for at least one month. One of the symptoms must be either 1, 2 or 3:
 1. Hallucinations
 2. Delusions (can be bizarre or non-bizarre)
 3. Disorganized speech (e.g., frequent derailment or incoherence)
 4. Grossly disorganized or catatonic behaviour
 5. Negative symptoms (e.g., affective flattening, alogia or avolition)
- Continuous disturbance for 6 months
- Social or occupational dysfunction (or both) for a significant portion of the time

DEFINITION

- Delusions:** False, unshared, and unshakable beliefs that persist despite overwhelming evidence to the contrary.
- Hallucinations:** Sensory perceptions without external stimuli from the relevant sensory organ. Auditory hallucinations are especially characteristic of schizophrenia but hallucinations in other modalities can also occur.

EXCLUDE POSSIBLE ORGANIC / MEDICAL CAUSES

- Endocrine, e.g., hyperthyroidism, hypoglycaemia
- Immunological condition, e.g., systemic lupus erythematosus (especially in females)
- Central Nervous System Disorders: tumours / Alzheimer's dementia / epilepsy / head trauma
- Drug (e.g., metamphetamine / amphetamine / **cannabis**) or alcohol abuse **or withdrawal**

Note: An organic cause should be actively investigated if:

- a) There is disorientation or memory loss.
- b) **There is a history of head trauma or anything suggestive of the issues above.**
- c) **There are abnormalities in vital signs, e.g., tachycardia, fever.**
- d) Hallucinations in modalities other than auditory are prominent e.g., knocking, water dripping.
- e) **The onset is very rapid within a few hours to days (beware of NMDA encephalitis, commonly in young ladies with ovarian teratoma, potentially lethal).**

EXCLUDE POSSIBLE PSYCHIATRIC CONDITIONS

1. Brief Psychotic Disorder

Presence of one or more of the following symptoms (delusions, hallucinations, disorganized speech, or grossly disorganised behaviour) for **at least a day to less than a month**.

2. Schizophreniform Disorder

Presence of one or more of the following symptoms (delusions, hallucinations, disorganized speech, or grossly disorganised behaviour) for **at least a month to less than 6 months**.

3. Delusional Disorder

Presence of non-bizarre delusions (involving situations that occur in real life such as being followed, poisoned, infected, loved from a distance) of at least a month's duration.

4. Mood Disorder

Symptoms and signs of a mood disorder must be brief relative to the duration of the symptoms and signs of schizophrenia. Otherwise, a diagnosis of schizoaffective or mood disorder with psychotic features may be more appropriate.

5. Schizoaffective Disorder

- An uninterrupted period of illness during which there is a major mood episode (major depressive or manic) concurrent with Criterion A of schizophrenia (delusions, hallucinations, disorganized speech, grossly disorganized or catatonic behaviour, negative symptoms).
- Delusions or hallucinations for 2 or more weeks in the absence of a major mood episode (depressive or manic) during the lifetime duration of the illness.

DISPOSITION AND LAW

- Cases with new-onset acute psychosis should be referred for a workup for organic causes discussed previously. If symptoms are mild to moderate, a fast-track referral to ASCAT can be made. If symptoms are severe, patients should be referred to the A&E for admission for organic workup.
- If patients with known schizophrenia present with a relapse, a suicide / homicide risk assessment should be performed. The Mental Health (Care and Treatment) Act (MHCTA) enacted 2008: only IMH is a designated psychiatric institution to enact this act.
 - i. Under section 7 of the Act, the police have the power (and in fact, a duty) to apprehend a person believed to be dangerous to himself or other persons and such danger is reasonably suspected to be attributable to a mental disorder, he/she may be apprehended by the police and taken for psychiatric examination.
 - ii. Under section 9 of the Act, if a medical practitioner believes a person to be mentally disordered or to require psychiatric treatment, the practitioner may send the person to a designated medical practitioner at IMH for treatment.
 - iii. Under section 10 of the Act, if a designated medical practitioner at IMH examines the person and is of the opinion that he should be treated as an inpatient at IMH, the practitioner may sign an order in accordance with Form 1 of the Schedule of the Act, pursuant to which, the person may be detained for up to 72 hours. This period may be extended for a further period of up to one month (pursuant to an order in accordance with Form 2 of the Schedule), and thereafter for a period no longer than six months (pursuant to an order in accordance with Form 3 of the Schedule).
 - iv. Under section 10(6) of the Act, a person may only be detained at IMH for treatment if (a) he is suffering from a mental disorder which warrants detention in a psychiatric institution for treatment; and (b) the detention is necessary in the interests of the health or safety of the person or for the protection of other persons.
 - v. The term “mental disorder” is broadly defined in section 2 of the Act as “*any mental illness or any other disorder or disability of the mind*”. (It is noted that while there were concerns raised about the broad definition, it appears to be the Parliament’s intention that the term be framed broadly, so that it may cover a wide range of mental illnesses.³ Further, in line with the purpose of the Act to only detain persons who require institutionalised treatment, whether the person requires such treatment should be assessed and determined by the examining professional on a case-by-case basis and not by legal definition.⁴)

MANAGEMENT

Medical / Bio

- Antipsychotic medications are the cornerstone of treatment (helps positive symptoms, but negative symptoms may remain e.g., cognitive difficulties, apathy, blunted affect).
- Patients should not be weaned off antipsychotics as the risk of relapse is very high, and switching of agents is not recommended unless there are severe side effects with current treatment.
- **For patients discharged from psychiatry SOC's, it is advisable to maintain at the same dosage unless there are side effects or need to reduce dosage in the elderly patients.**
- Risk of relapse without medication: 77% within 1 year, more than 90% beyond 2 years ².
- Of those treated, almost half are stable enough to be in the community.
- General principles of drug treatment: maintenance dose should not be less than half the effective dose at acute phase. Trial medications at adequate dose and duration of at least 4-6 weeks before switching agents.
- If this is a case of relapsed schizophrenia and unlikely organic cause (e.g., strong genetic loading), and risk to self / others is low, and has suitable caregiver:
 - ➔ Perform investigations: FBC / renal panel / LFT / TFT / ESR (add lipid / glucose if there are risk factors and for baseline)
 - ➔ Start risperidone 1 mg ON
 - ➔ Refer direct access psychiatry according to zoning principles.

Note: For stable known cases of schizophrenia not known to any public psychiatrists yet, can refer to the Early Psychosis Intervention Programme (EPIP) @ IMH – they will follow up those with prodrome / at risk mental state for 3 years, and referral is via Epic directly to EPIP.

Metabolic Syndrome in Patients with Schizophrenia

- Patients with schizophrenia have high risks of developing metabolic syndrome due to
 - o Sedentary lifestyle
 - o Smoking
 - o Side effects of antipsychotics
 - o Other mental health comorbid of intellectual blunting, fixed beliefs, and ideas
- As family physicians, we will need to monitor fasting lipids, glucose, and weight annually. ³
- Encourage healthy diet and regular exercise.
- Reduce smoking.

Psychosocial

In the stable phase of the illness, psychosocial interventions can help patients recover independence.

- Individual therapy e.g., Cognitive Behavioural Therapy if there is good insight.
- Family therapy
- Social skills training
- Vocational rehabilitation
- A good place to refer is the Anglican Care Centre (COMIT team) via NUP MSW. Services provided include:
 - o Counselling, case management
 - o Daycare centre, caregiver support
 - o Integrated employment service
 - o Sheltered home for patients with mental health issues

ANTIPSYCHOTICS

Oral Antipsychotics

Drug	Usual Maintenance Dose	Maximum Dose	Remarks
First-Generation / Typical Antipsychotics (FGA)			
Haloperidol (S1) \$^A	Moderate symptomology: 1.5mg BD or TDS Severe symptomology: 3-5mg BD or TDS	30mg/day	
Trifluoperazine (S1) \$^A	15-20mg in divided doses	40mg/day	
Chlorpromazine (S1) \$^A	Moderate symptomology: 100mg TDS or QDS Severe symptomology: 250mg TDS	800-1000mg/day	Common ADR: sedation, anticholinergic effects, postural hypotension
Sulpiride (S2) \$\$\$\$^A	Moderate symptomology: PO 400mg BD to QDS Severe symptomology: PO 2400mg/day in 2-4 divided doses	2400mg/day	
Thioridazine (Not available in NUP)	600-800mg/day, divided in 2-4 doses	800mg/day	Common ADR: sedation, anticholinergic effects, postural hypotension
Second-Generation / Atypical Antipsychotics (SGA)			
Quetiapine (S2) \$\$\$\$^A	Immediate-release tablets: 300mg ON Extended-release tablets: 400-800mg OD (Not available in NUP)	800mg/day	Common ADR: sedation, anticholinergic effects, postural hypotension Given at night due to sedation
Risperidone (S2) \$\$\$\$^A	2-6mg OD	8mg/day	Highest risk among SGA for: - Hyperprolactinemia - EPSE (especially at doses > 4mg/day)
Olanzapine (S2) \$\$\$\$^A	10-20mg OD	30mg/day	Common ADR: sedation, anticholinergic effects, significant weight gain
Aripiprazole (Not available in NUP)	10-15mg/day	30mg/day	

Drug	Usual Maintenance Dose	Maximum Dose	Remarks
Clozapine (Not available in NUP)	300-450mg/day, in divided doses	900mg/day	Managed by psychiatrists Common ADR: sedation, anticholinergic effects, weight gain Serious ADR: - Agranulocytosis, occurs in about 0.34% of patients (greater risks in females or older age). Requires weekly FBC for 18 weeks, then monthly thereafter as long as they're on clozapine. - Myocarditis and cardiomyopathy - Lowered seizure threshold
Paliperidone (Not available in NUP)	6-12mg/day	12mg/day	

^ Cost \$ <\$10 per month \$\$ \$10 - <\$20 per month \$\$\$ \$20 - <\$30 per month
 \$\$\$\$ \$30 - <\$60 per month \$\$\$\$\$ \$60 - <\$90 per month \$\$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS).

Unit prices before GST and accurate as of Apr 2024.

Prices shown are an estimate. Please contact pharmacy for accurate pricing.

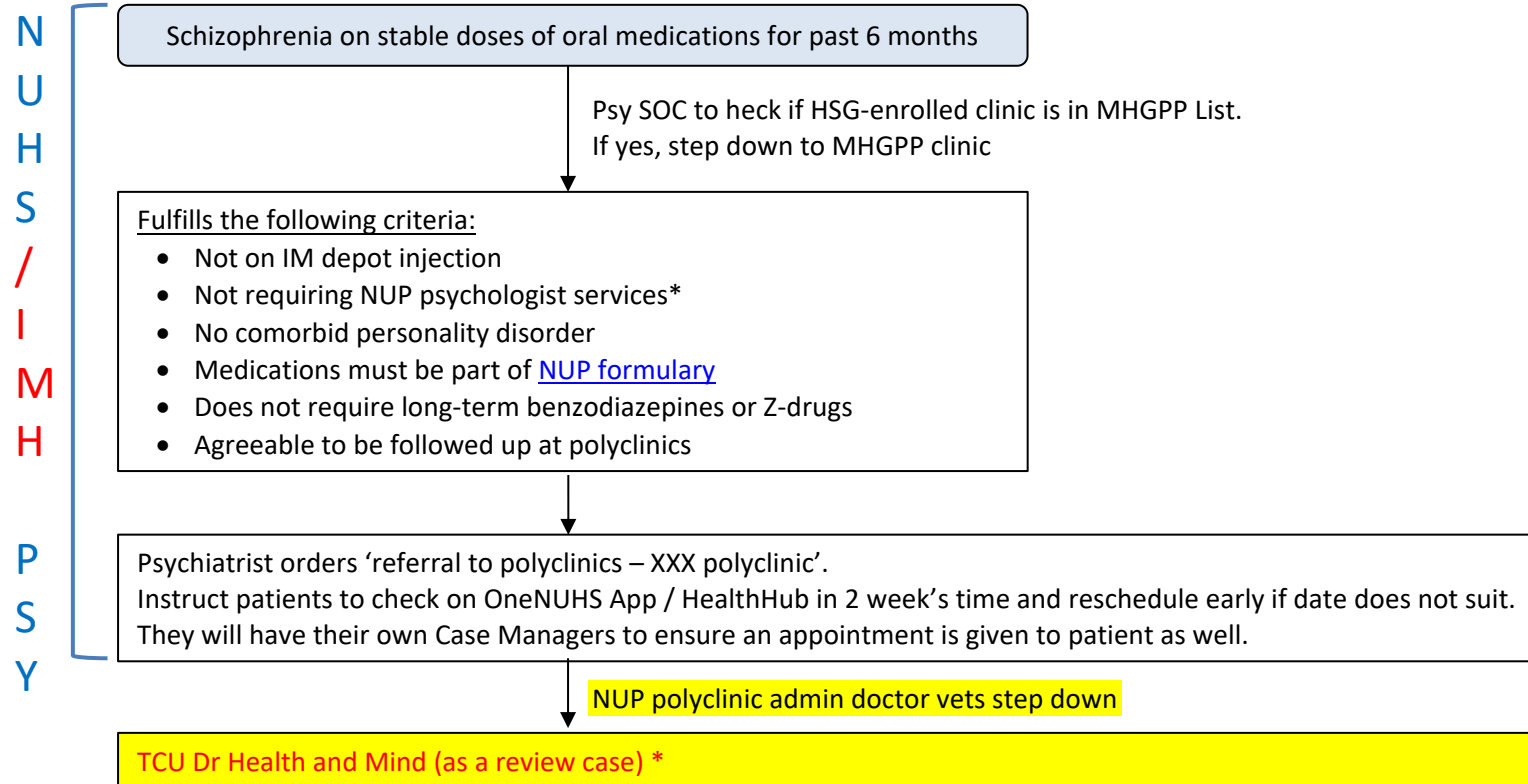
Depot (IM) Antipsychotics (Not available in NUP)

- FGA
 - Zuclopenthixol (e.g., Clopixol, for aggressive patients)
 - Flupenthixol (e.g., Fluanxol, for those with depression)
 - Fluphenazine (e.g., Modecate)
 - Pipotiazine (e.g., Piportil)
 - Haloperidol
- SGA
 - Paliperidone

Adverse Drug Reactions (ADR) of Antipsychotics

ADR		Remarks
Extrapyramidal (Less common in SGA, lowest risk with quetiapine, clozapine)	Parkinsonism	Non-urgent Can be relieved by benzhexol 2-4mg daily
	Acute dystonia	Urgent Refer to Emergency Department for IM benztropine Reduce dose of antipsychotic
	Akathisia	Urgent Refer direct access psychiatry Call psychiatrist on call to reduce dosage of antipsychotics whilst awaiting appointment Clonazepam BD PRN or propranolol 10-20mg TDS PRN
	Tardive dyskinesia	Consider vitamin E 400-1600IU/day or melatonin 10mg/day Consider referral to specialist if bothersome or at risk of falls Consider SGA or refer to neurologist for tetrabenazine
Neuroleptic malignant syndrome	Characterized by hyperthermia, muscular rigidity, autonomic instability, altered conscious level	Emergency, refer ED Can be fatal in up to 20% of cases Tends to happen in those taking high dose FGA
Cardiovascular	ECG changes, e.g., widened QRS complex, prolonged QT interval, tachycardia, arrhythmias	Emergency For prolonged QTc >500ms, refer Psychiatry SOC direct access for dose adjustment or switching to aripiprazole
	Postural hypotension	Initiate at lower dose and slow titration in older or frail patients
Hyperprolactinemia	Menstrual irregularities, infertility, galactorrhoea, sexual dysfunction	Refer to specialist Consider aripiprazole
Anticholinergic	Dry mouth and eyes, blurred vision, constipation, urinary retention	Higher risk in older patients
Antihistaminic	Sedation	For medications with high risk of sedation, take at night and titrate slowly Most severe during first few weeks of treatment
Seizures	Lower seizure threshold in at-risk individuals	Caution in patients at risk for seizures or taking other medications that lower seizure threshold
Metabolic syndrome	Weight gain, diabetes, dyslipidaemia	Monitor weight, blood pressure, fasting glucose, lipid profile
Hematologic	Agranulocytosis, neutropenia, leukopenia	Monitoring recommended if previously experienced drug-induced leukocytopenia or have preexisting low white blood cell or absolute neutrophil count Common in clozapine, routine monitoring recommended

STEPDOWN STABLE SCHIZOPHRENIA WORKFLOW



**If there are ongoing psychologist appointments in NUHS / IMH Psychiatry department, that arrangement is to continue.*

Sample of Patient Information leaflet patients may obtain from NUHS Psychiatry, which would include NUH clinic number, NUH Emergency Department manning hours by psychiatrist, IMH hotline:

Action Plan to Prepare for Schizophrenia Relapse

Use this sheet to prepare yourself or your loved ones to help you with a psychotic episode. Your physician can help you to fill up the following details:

	Contact details
Healthcare provider	
Friends and family	
Emergency resources	

Identifying relapse

Signs of relapse varies from person to person. Early warning signs may include:

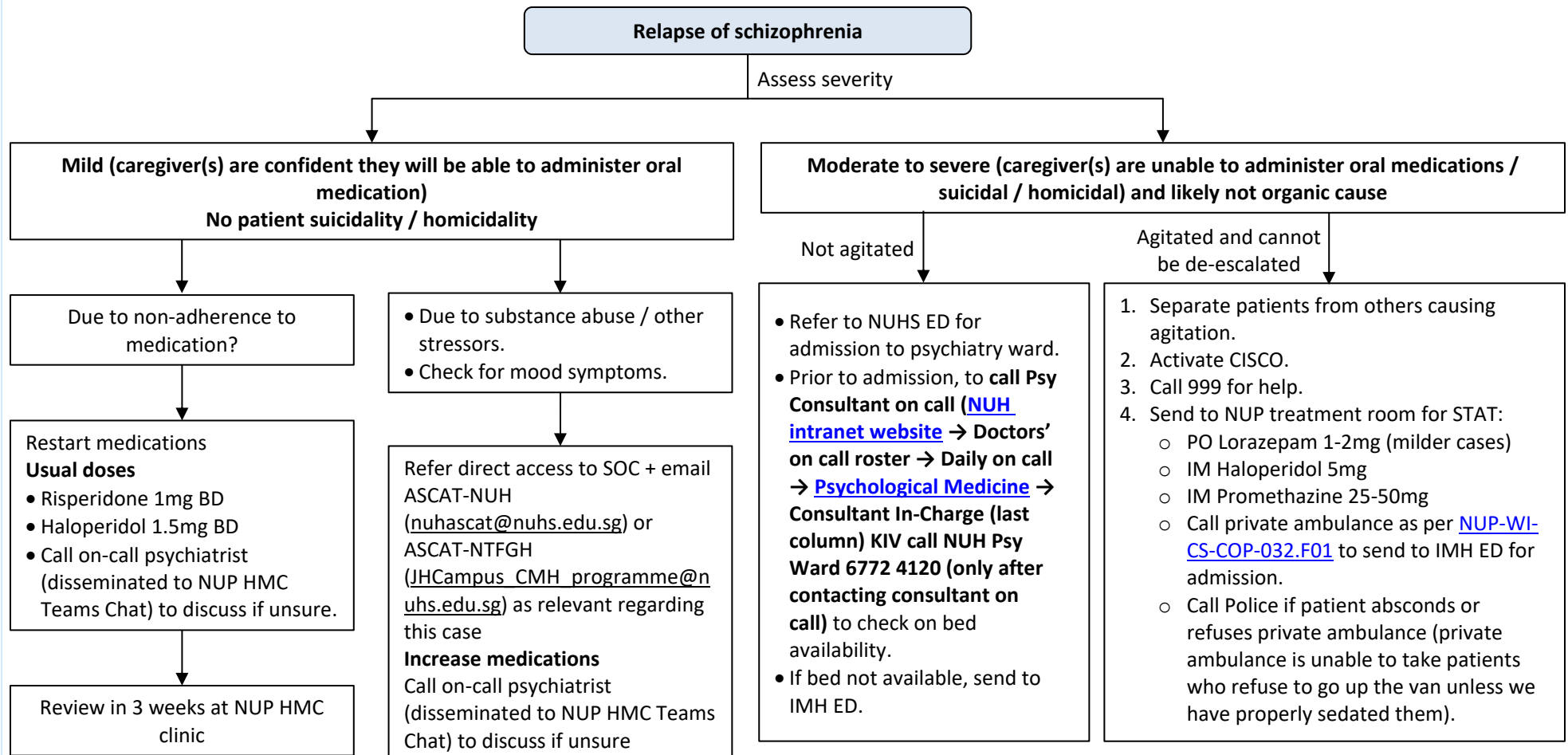
- Difficulty concentrating
- Feeling suspicious or paranoid around others
- Spending more time alone
- Speaking in a confused manner
- Difficulty telling reality and fantasy apart
- Suddenly doing worse at work or school

A relapse may present similarly to your previous episodes. Take a moment to reflect on the signs which you noticed during your previous relapse (if any) and write it down in the box below:

During my previous relapse, the signs that I observed were	
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WHAT TO DO IF PATIENT RELAPSES

- No clear set of criteria that defines relapse.⁵
- Relapse signature varies for each patient, might be similar to previous presentation, and usually symptoms return gradually.
- Some relapses occur without warning, but some may have early signs such as insomnia, social withdrawal, difficulty concentrating, loss of interest, increasing paranoia and hallucinations.
- Factors that may drive relapse e.g., non-adherence to medication, stressors, substance abuse.^{6,7}
- Workflow below for reference; doctors from general pool to escalate to Clinic Head / Deputy / HMC Doctor-in-charge within the clinic.



WHAT TO DO WHEN SEEING PATIENTS WITH STABLE SCHIZOPHRENIA

Recommended Care Components	Minimum Frequency	Remarks
Monitoring for metabolic side effects (e.g., fasting glucose and lipid), check BMI	Annually	For patients on typical AND atypical antipsychotics
Comorbid anxiety and depression	Annually	If suspected
Global Assessment of Functioning (GAF) score	Annually	Check functional status at home and in the community
Dentition	Annually	Refer NUP Geriatric and Special Needs Service (NUP GSND) at QTP if no primary dentist, and dependent on caregiver for personal hygiene
Clinical review	At least twice a year	Assess symptoms, response and adherence to medications, side effects
Psychosocial interventions	Annually	TCU MSW for Anglican Care Centre if patient keen to go

RESOURCES

Resources	Remarks
Community Intervention Team (COMIT) Provides counselling and psychoeducation support for patients with mental health issues and their caregivers Free of charge	Requires a referral to MSW. Alternatively go to mindline.sg and can self-refer to COMIT.
Early Psychosis Intervention Programme (IMH) Early detection and treatment of mental disorders among those age 18 to 40	Can refer directly to EPIP on Epic. Contact number: 6389 2972
Family Service Centers (FSCs)	To find the nearest FSC: https://www.msf.gov.sg/our-services/directories
Samaritans of Singapore (SOS) hotline	1767 (24 hours)
Singapore Association for Mental Health (SAMH)	Contact number: 6255 3222 Email: comms@samhealth.org.sg

ROLES OF HEALTHCARE TEAM

1. Admin Doctors

- Ensure step downs from NUHS Psychiatry SOC's for stable schizophrenia's are right sited as "TCU Dr HMC (as a review case)".
- Does not need CM pairing for step down care.

2. HMC Doctors

- Ensure patient is stable, adherent to medications.
- Do mental state examination.
- Be a point of contact / means of liaison with psychiatry specialists as required for cases of mild relapse.
- Ensure patient's metabolic health is taken care of during consultation.

3. Non-HMC Doctors

- Escalate as necessary for patients with mild relapse.
- TCU Psychologist if patient has concomitant mood or anxiety that is not because of schizophrenia, ensure patient is alert enough and suitable for CBT. Discuss with psychologist if unsure.

4. MSW

- Ensure patient is supported socially and is integrated into the community.
- Point of referral to COMIT's / SAMH / Anglican Care Centre.

5. Psychologist

- Conduct psychotherapy as necessary for suitable patients with other comorbidities such as anxiety, depression, insomnia.

6. Care Manager

- TCU CM within relevant teamlet, for chronic care management as per usual DM / HTN / HLD workflows.

REFERENCES

1. Subramaniam M, Abidin E, Vaingankar JA, Sambasivam R, Zhang YJ, Shafie S, Basu S, Chan CT, Tan CS, Verma SK, Tang C, Chua HC, Heng D, Chong SA. Lifetime Prevalence and Correlates of Schizophrenia and Other Psychotic Disorders in Singapore. *Front Psychiatry*. 2021 Mar 11;12:650674. doi: 10.3389/fpsyt.2021.650674. PMID: 33776823; PMCID: PMC7991584.
2. Zipursky RB, Menezes NM, Streiner DL. Risk of symptom recurrence with medication discontinuation in first-episode psychosis: a systematic review. *Schizophr Res*. 2014 Feb;152(2-3):408-14. doi: 10.1016/j.schres.2013.08.001. Epub 2013 Aug 21. PMID: 23972821.
3. Chronic Disease Management Programme – Handbook for Healthcare Professionals. Ministry of Health Singapore (2021)
4. Sweeney M, Whiskey E, Patel RK, Tracy DK, Shergill SS, Plymen CM. Understanding and managing cardiac side-effects of second-generation antipsychotics in the treatment of schizophrenia. *BJPsych Advances*. 2020;26(1):26-40. doi:10.1192/bja.2019.49Siafis S, Brandt L, McCutcheon RA, Gutwinski S, Schneider-Thoma J, Bighelli I et al. Relapse in clinically stable adult patients with schizophrenia or schizoaffective disorder: evidence-based criteria derived by equipercntile linking and diagnostic test accuracy meta-analysis. *The Lancet, Psychiatry*. 2023 11(1): 36-46. Nov. doi: 10.1016/S2215-0366(23)00364-4.
5. Siafis S, Brandt L, McCutcheon RA, Gutwinski S, Schneider-Thoma J, Bighelli I et al. Relapse in clinically stable adult patients with schizophrenia or schizoaffective disorder: evidence-based criteria derived by equipercntile linking and diagnostic test accuracy meta-analysis. *The Lancet, Psychiatry*. 2023 11(1): 36-46. Nov. doi: 10.1016/S2215-0366(23)00364-4.
6. Almuqrin A, Georgiades A, Mouhitzadeh K, Rubinic P, Mechelli A, Tognin S. The association between psychosocial stress, interpersonal sensitivity, social withdrawal and psychosis relapse: a systematic review. *Schizophrenia (Heidelb)*. 2023 Apr 10;9(1):22. doi: 10.1038/s41537-023-00349-w. PMID: 37037858; PMCID: PMC10086007.
7. Masroor A, Khorochkov A, Prieto J, Singh KB, Nnadozie MC, Abdal M, Shrestha N, Abe RAM, Mohammed L. Unraveling the Association Between Schizophrenia and Substance Use Disorder-Predictors, Mechanisms and Treatment Modifications: A Systematic Review. *Cureus*. 2021 Jul 29;13(7):e16722. doi: 10.7759/cureus.16722. PMID: 34513357; PMCID: PMC8405179.